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Recent giant detachment of a glacier on the Tibetan plateau provoked by its frozen tongue

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Sudden detachments of entire glacier tongues are rare but potentially devastating events that remain poorly understood. Here, we report on an event that occurred in Tibet on November 1, 2022, with an ice volume of ~40 million cubic meters, ranking it among the largest recorded ice avalanches. Utilizing a broad array of satellite data and seismic signals, we reconstruct the motion of the source glacier leading to its failure and the large landscape changes caused by the avalanche. The lack of glacier advance during its acceleration phase suggests that a frozen terminus has played a key role in the evolution of the glacier failure by acting as a dam to the ice mass pushing from above and its potential water content. The dense coverage by satellite data reveals details of an exponential increase in glacier speeds, up to 46 m per day in the weeks and days prior to failure.

Flow instabilities and failure can turn sluggish glaciers into violent hazards. Incorporation of sediments and rocks, along with the frictional production of meltwater, can transform ice into highly mobile mass flows[1–3]. Typical ice avalanches occur when ice breaks off steep glacier fronts or when sections of steep glaciers, generally exceeding 30° in inclination, fail[4]. Glacier surges represent a second type of glacier instability, characterized by reduction in basal friction over weeks to few years, allowing glaciers to flow orders of magnitude faster than during their quiescent phases[5–7]. Such surges can cause long glacier advances of up to several kilometres in just a few weeks to months.

A third type of glacier instability involves catastrophic detachments of entire low-angle tongues, usually not steeper than 20°. This process was first detailed internationally for Kolka Glacier, Caucasus. In 2002, its tongue suddenly collapsed, creating a $130 \times 10^6\,m^3$ ice-rock avalanche that reached speeds of up to ~$300\,km\,h^{-1}$, devastating the Karmadon village after an 18 km runout and transforming further into a 15 km long mudflow, resulting in approximately 120 fatalities[8–11]. Initially considered a unique disaster, this type of event was later echoed by two massive glacier detachments in July and September 2016, when large parts of two neighbouring low-angle glaciers in the Aru range, western Tibet, suddenly failed, resulting in ice-rock avalanches of 68 and $83 \times 10^6\,m^3$, respectively, and killing nine herders and hundreds of livestock along their 5–6 km long runout[12]. The Aru twin glacier collapses spurred broader interest in the surprising detachments of low-angle glaciers and instigated further research that identified a dozen comparable events worldwide in recent decades across Eurasia, North- and South America[11]. Another notable event was the $130 \times 10^6\,m^3$ detachment of Sedongpu Glacier, southeast Tibet (Fig. 1a), which completely dammed the Yarlung Tsangpo/Brahmaputra River for several days in 2018[11,13].

Avalanches consisting of ice or ice-rock mixtures, such as the devastating 28 May 2025 Nesthorn/Birch Glacier/Blatten avalanche in the Lötschen valley, Switzerland, can be particularly mobile, leading to strongly increased runout distances compared to rapid rock mass movements without ice or snow components[3,14–16]. Factors leading to such far avalanche reaches include wet and dry basal lubrication, fluidization within the entire volume, through processes such as ice melt due to frictional heating and particle collisions, incorporation of water along the avalanche path, and mechanical crushing of basal ice[16–19]. Apparent friction coefficients tend to reduce with increasing avalanche volumes[16]. Liquid water embedded in the glacier and sediments before failure can amplify the avalanche mobility. Also, ice and snow surfaces, in cases where the avalanche travels over those, can reduce basal friction. The lowest angles of reach (so-called Fahrböschung) for ice-rock avalanches are found for the few ones resulting from glacier detachments[11], but reasons for their particularly low friction are hardly understood; particularly soft and wet sediments underneath the detaching glaciers, accumulation of water under high pressure, and the particularly large volumes involved are so far believed to play a role[8,11,12,20].

Every new glacier detachment offers thus a possibility to reduce the large knowledge deficits related to this rare but potentially devastating type of mountain hazards. Here we describe and analyse a previously unnoticed large glacier collapse in eastern Tibet, which provides indeed important new

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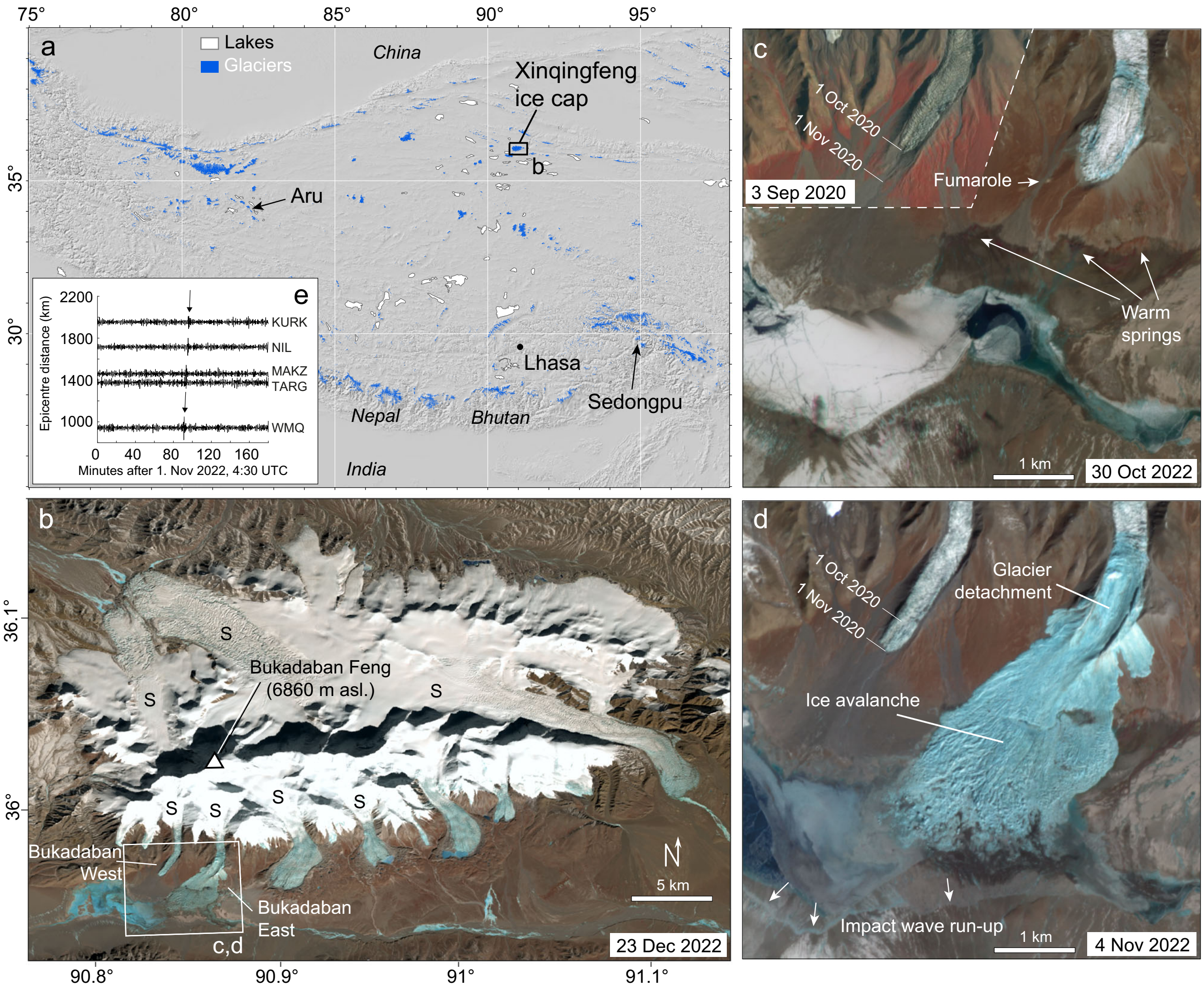

Fig. 1 | The massive ~40 million cubic meter glacier avalanche of 1 November 2022 from the Xinqingfeng ice cap, Kunlun mountains, Tibetan plateau. **a** Location of the Xinqingfeng ice cap and two other previous glacier collapses, Aru 2016 and Sedongpu 2018. **b** Landsat 9 satellite image of Xinqingfeng ice cap with its highest peak, Bukadaban Feng. Glaciers with known surge activity are indicated by letter S. The rectangle indicates location of (**c**, **d**). **c** PlanetScope satellite image 2 days before detachment. For the Bukadaban West glacier, a PlanetScope image of September 2020 is overlaid, just before its sudden and rapid surge advance by 500 m. Locations of warm springs are indicated. Large parts of the lake in the valley bottom are already ice-covered. **d** PlanetScope satellite image 3 days after detachment. The detached glacier tongue [90.869° E, 35.969° N] had 1.1 km horizontal length, an elevation range of ~5010–5250 m asl, an area of ~0.5 km^2, and 12° surface slope. The ice avalanche caused an impact wave that ran up to 800 m across the lake shore. Panels **a**, **b** in plate carrée, and **c**, **d** in UTM zone 46 projection. **e** Seismic signals recorded at stations surrounding eastern Tibet (located outside map area, Supplementary Fig. S2) over the time window during which the avalanche must have happened according to satellite images. Station names are given as 3–4 letter IRIS station codes. The only significant event signal suggests a detachment time of roughly 5:55 UTC (13:55 Bejing time). Topographic hillshade in **a** is derived from the Shuttle Radar Topography Mission elevation model (data courtesy of the U.S. Geological Survey); Landsat image courtesy of the U.S. Geological Survey[72]; PlanetScope images courtesy of Planet[73].

insights into the mechanisms involved in the detachments of low-angle glaciers.

Results

Reconstruction of the unnoticed avalanche and its giant volume

Through a routine worldwide mapping of surging glaciers[21,22] (Supplementary Fig. S1), we discovered massive ice avalanche deposits on the southern flank of the Xinqingfeng ice cap (Fig. 1a–d). Satellite imagery revealed that the ice avalanche occurred on 1 Nov 2022 between 4:35 UTC and 6:54 UTC, with nearby seismic stations indicating an approximate event time of 5:57 UTC (13:57 Beijing time; Fig. 1e and Supplementary Figs. S2–S5, "Methods"). The event location is part of Hoh Xil Nature Reserve, with restricted access and no local population, and the event was thus to our best knowledge, not reported previously. Its remote location, the very complicated access, and the fact that we discovered the avalanche only two years after its occurrence, require the event analysis to be solely based on remote sensing data and leave inevitably some uncertainties that timely in-situ data might have helped to clarify.

The 1 Nov 2022 giant ice avalanche was caused by the detachment of the lower part (1.1 km length; 12.3° glacier surface slope) of an unnamed glacier, which we here refer to as Bukadaban East (Fig. 1c, d). The avalanche descended to a lake at ~4880 m asl., covering an area of 5.3 km^2. Measured from the upper detachment scarp to the furthest deposit point (elevation drop 370 m, horizontal distance 4300 m), the event displayed a remarkably low overall slope angle of 4.9°, even less than the 2016 Aru ice avalanches[12]. Simulations using a numerical avalanche model suggest an average friction coefficient of the avalanche of 0.11 before reaching the lake, very similar to the values suggested for the Aru collapses[12,23] (0.10–0.14), and indicate avalanche speeds of up to ~200 km h^{-1} ("Methods"). Remarkably, an exceptionally low initial Coulomb (dry) friction of 0.05 is necessary to let the

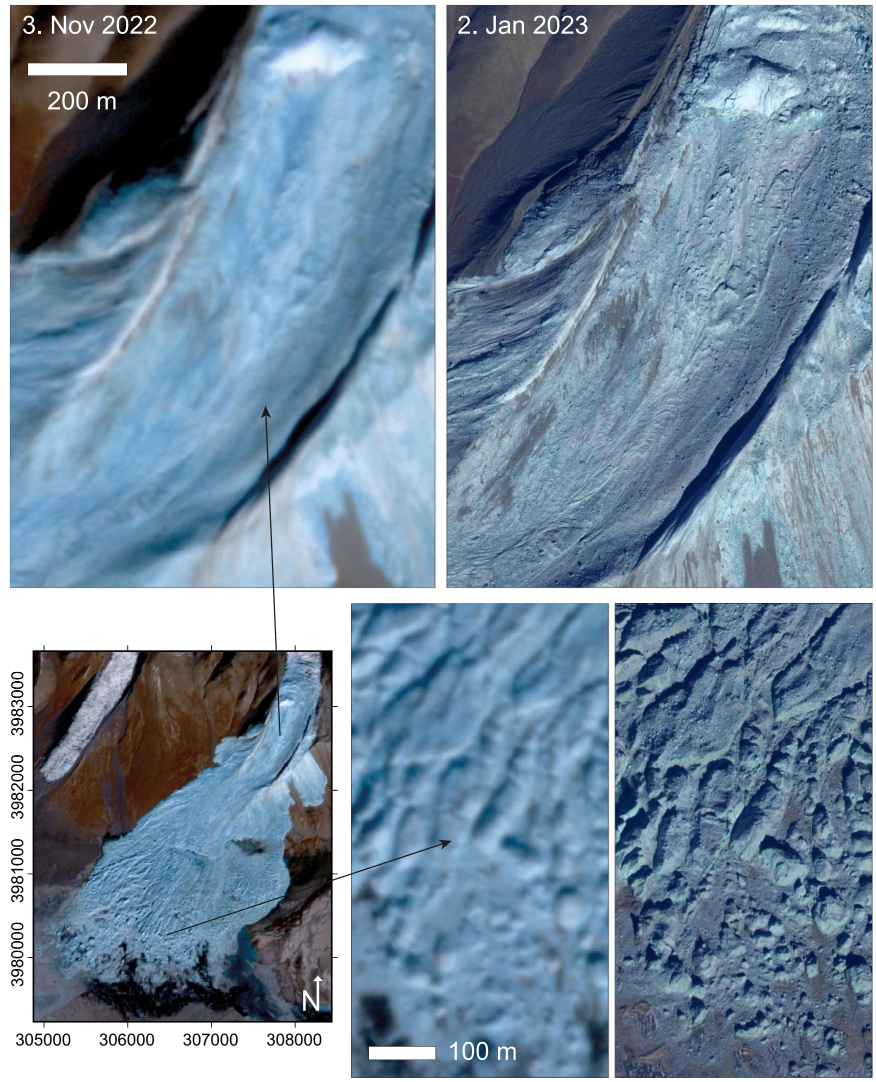

Fig. 2 | Details of satellite images. Details of detachment and avalanche deposits in satellite images of 3 Nov 2022 (2 days after detachment; PlanetScope, 3 m spatial resolution) and 2 Jan 2023 (2 months after detachment; WorldView, 2 m resolution multispectral sharpened with 0.5 m pan data). A PlanetScope image of 2 Nov 2022 (22 h after detachment; not shown due to some thin clouds) does not show any differences to the 3 Nov image, except some small melt-related changes at the locations of warm springs at the slope foot. Coordinates are in UTM zone 46. WorldView data © Vantor, PlanetScope images courtesy of Planet[73].

avalanche simulation match the observed extent. The avalanche travelled 1.7 km in the lake, generating an impact wave that ran 700–800 m across the flat opposite lake shore with a run-up height of up to 13 m above lake level (Fig. 1d).

Independent seismic inversion of the event, based only on the seismic signals recorded at stations ~900–1900 km away from the epicentre, and assuming one synchronous sliding constant mass block, suggests maximum speeds of ~12 $m\ s^{-1}$ (43 $km\ h^{-1}$), southward movement of the mass for the first 53 s, and then a turn to southwest for another 90 s ("Methods"). Given the large uncertainty associated with the very distant and weak seismic recordings, these results, in particular the avalanche direction, are roughly consistent with the numerical avalanche modelling and other findings about the avalanche (Methods).

Post-detachment satellite images reveal only clean ice on both the detachment surface and the ice avalanche deposits (Figs. 1d and 2), unlike many other low-angle glacier avalanches, which obviously involved fine lithospheric sediments from the glacier bed[11]. The detachment surface exhibits fine-grained ice debris (Fig. 2). We can, however, not rule out that remains of the avalanching glacier covered the glacier bed and possible fine sediments only after the initial failure. Preservation of vegetation patches and microtopography, such as stream channels or small ridges and depressions, and the lack of terrain elevation changes found in ICESat-2 elevations (Supplementary Figs. S6–S9), suggest minimal erosion of the underlying terrain along the ~5° steep, ~900 m long slope between the former glacier tongue and the lake plain. This observation aligns with the modelled existence of continuously frozen ground around the glacier[24], with mean annual ground surface temperatures of about −3 °C ("Methods"). In the lake area, where most avalanche ice was deposited, up to 15–20 m thick, many large intact blocks of glacier ice are visible in high-resolution post-event satellite images (Fig. 2), suggesting they had a smooth ride, without strong forces from friction and shearing.

At the slope base, satellite imagery indicates a fumarole, with a smoke fan visible in both pre-event and post-event high-resolution data, and warm springs, or at least enhanced geothermal heat-flow in combination with ponds and water sources[25] (Figs. 1c and 3). Also, the excavation of new lake area in this zone by the avalanche points to non-frozen soft sediments. In contrast to the slope above the valley floor, substantial erosion by the avalanche happened in the valley floor as suggested by the landscape changes caused by the avalanche and illustrated by our avalanche model run (Fig. 4; "Methods"; Supplementary Fig. S10). We found no signs of strongly enhanced geothermal activity at the detachment site itself (Figs. 2 and 3).

We estimated the avalanche volume using stereo processing of high-resolution optical satellite images (Fig. 5), glacier thickness models[26,27], ICESat-2 laser altimeter tracks (Supplementary Fig. S6), and medium resolution ASTER satellite stereo images. Combined, these data suggest a detached volume of approximately 40×10^6 m^3, with a conservatively estimated uncertainty of roughly $\pm 5 \times 10^6$ m^3. The estimated detachment volume is equivalent to an average detached ice thickness of around 80 m ("Methods"). These numbers rank the event among the largest ice

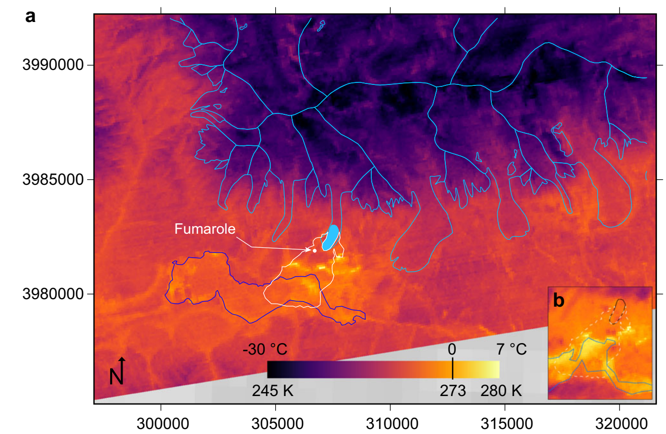

Fig. 3 | Night-time thermal infrared scene. **a** Advanced Thermal Emission and Reflection Radiometer (ASTER) night-time thermal infrared image of 1 May 2021, processed to product level AST-08, kinetic surface temperature. Light blue outlines: GLIMS glacier outlines, light blue area: detached glacier section, white outline: ice avalanche deposits, dark blue outline: lake outline. Temperature unit of legend is Kelvin. Bright yellow zones are up to +6 °C, while the rest of the landscape has surface temperatures far below zero degrees Celsius. On the date of acquisition the lakes were frozen, but there was no snow cover on the surface outside of the glaciers. Coordinates are in UTM zone 46. **b** ASTER night-time thermal infrared of 20 August 2025 showing no enhanced temperature of the former detachment area (black outline). Maximum temperatures are 285 K. (Former avalanche area: white dashed line; lake area of Aug 2025: blue line). ASTER image data courtesy of NASA/METI/AIST/Japan Space Systems and U.S./Japan ASTER Science Team.

avalanches recorded to date. A secondary ice avalanche in October 2023 added another ~10×10^6 m^3 to the detached ice volume (Fig. 4). Additionally, we estimate that the ice avalanche eroded and transferred several million m^3 of sediments in and around the lake, strongly altering 3.5 km^2 of valley floor (Fig. 4, Methods).

Surge-like advance of the neighbouring twin glacier

Because the 2016 Aru twin glacier detachments happened within just two months, the western neighbour to the Bukadaban East glacier, termed here Bukadaban West, warrants interest as both share, as in the Aru case, similar shapes and settings. Over the period 2000–2018, both glaciers exhibited surge-like mass transfers with an elevation bulge of approximately 20 m or more moving down the glacier tongues, as indicated by elevation differences ("Methods," Supplementary Fig. S11). In Oct 2020, Bukadaban West suddenly advanced by 500 m, at nearly 17 m day^{-1}. In contrast, the Bukadaban East tongue showed minor length changes over the recent 50 years (Methods). Overall, since the 1970s, the glaciers of the Xinqingfeng ice cap have been nearly balanced with average specific mass losses of not more than 0.2 m w.e[28–31]. This moderate mass loss, in global comparison, is likely connected to increasing precipitation/snow fall, as indicated by strongly increasing endorheic lakes in the area[30–32] ("Methods"). Several glaciers of the ice cap have shown surge behaviour historically[21,28,33,34] (Fig. 1b).

Exponential acceleration of ice motion towards failure

Cross-correlation of repeat satellite images[35] (Fig. 6d, e) reveals that the Bukadaban East glacier tongue had background speeds of approximately 0.2 m day^{-1} (70 m year^{-1}) until June 2022. Notably, along the lowermost 500 m glacier length, there was no detectable movement within measurement accuracy. From July 2022 on, speeds increased dramatically, peaking at 26 m/day between 30 and 31 Oct, and 46 m day^{-1} between 31 Oct and 1 Nov. By mid-to-late 2022, speeds were highest and predominantly increased on the later detaching glacier part, suggesting that reduction in friction or weakening of the glacier base developed over 3–4 months prior to detachment. From approximately September 2022 on,

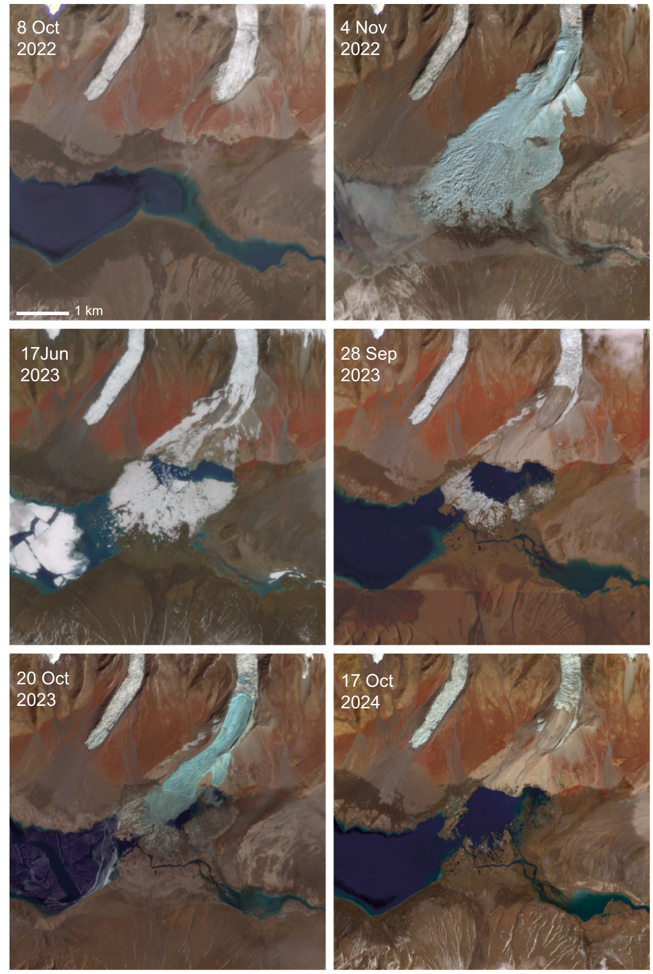

Fig. 4 | Satellite image series. Pre- and post-detachment series of PlanetScope satellite images of the Bukadaban West and East glacier tongues and the ice avalanche area showing important landscape changes caused by the avalanche. PlanetScope images courtesy of Planet[73].

the detaching glacier part became largely decoupled from the glacier above. In the days leading to detachment, the surface of the rapidly moving glacier tongue remained largely intact, though the margins increasingly comprised light-blue ice (Figs. 1c and 5c), which we interpret as wet slush ice and/or dry crushed ice ("Methods"). This development indicates primarily block movement of the lower glacier section. Notably, the terminus position remained close to unchanged despite the strong acceleration phase. In comparison, the Bukadaban West glacier also accelerated over several months, but it reached speeds of only up to 20 m day^{-1} before advancing its terminus by 500 m in 2020 (Fig. 6a).

The dense repeat coverage by high-quality satellite images enables the detailed measurement of an exponential acceleration phase during the weeks and days prior to failure. For the Aru glacier detachments in 2016, for instance, no suitable satellite data are available to reconstruct ice speeds during the 1–2 weeks before collapse. Before this final acceleration phase, though, Aru glacier ice speeds were roughly half as high as in the Bukadaban case, and we did not find signs of such a massive surface rupture as occurring on the Bukadaban East glacier.

According to climate reanalyses and the closest meteorological station data, the early acceleration phase of Bukadaban East glacier coincided with a warm weather period; June and July 2022 were notably warm, with May and August 2022 being exceptionally warm, nearly 4 °C above the 1991–2020 average. Overall, 2022 was nearly the warmest year for the study region and particularly wet (Fig. 7, "Methods").

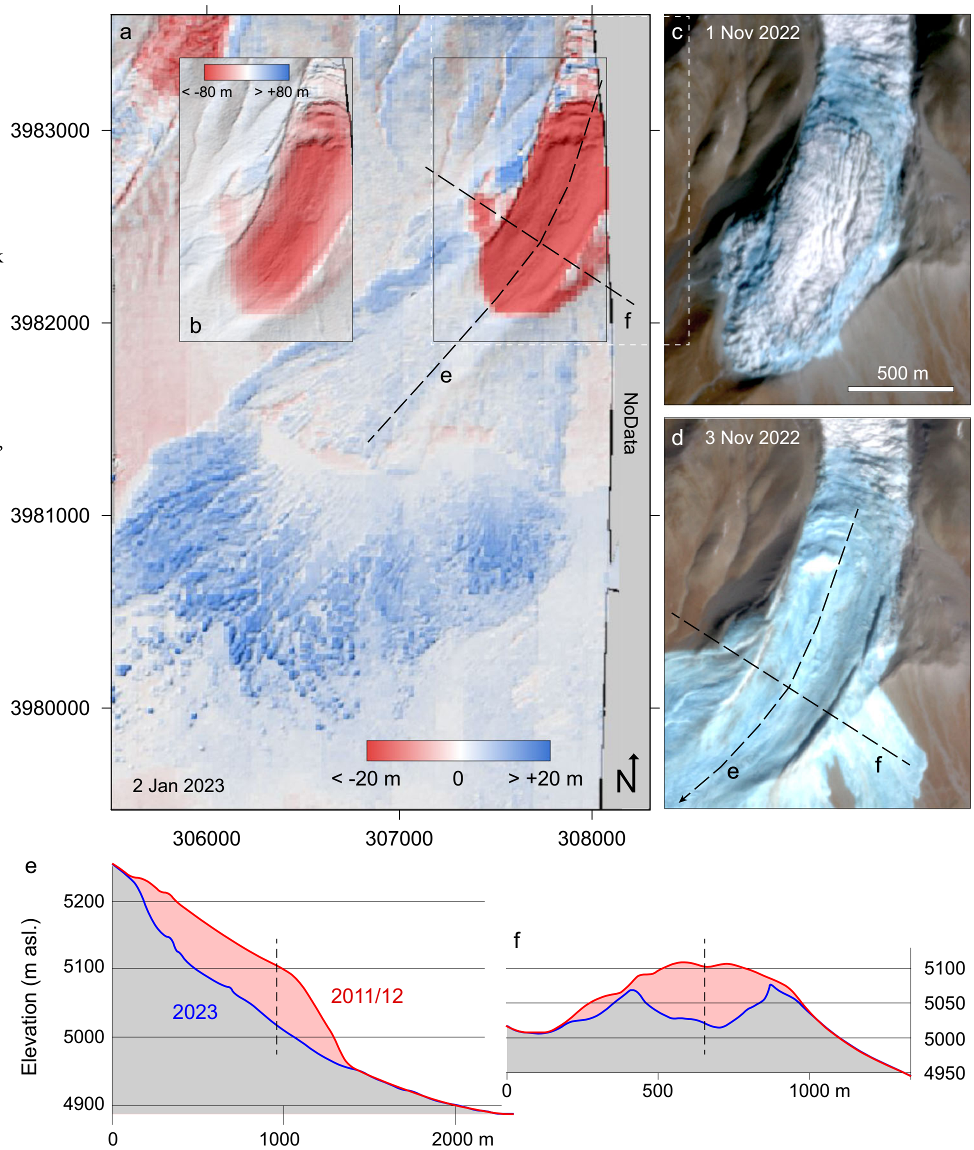

Fig. 5 | Elevation differences between 2011/2012 and 2 Jan 2023 over the detachment and ice avalanche deposits. **a** Elevation differences between a post-event digital elevation model derived from overlapping WorldView high-resolution optical satellite data of 2 Jan 2023 and the pre-event 2011/2012 Copernicus DEM. Color-coded elevation differences are transparently superimposed over a hillshade of the WorldView-derived DEM. Elevation changes are saturated at ±20 m. **b** same as black rectangle in (**a**), but with saturation of elevation differences at ±80 m. **c**, **d** PlanetScope satellite images of 1 Nov 2022, around 100 min before detachment, and 3 Nov 2022. The white dashed rectangle in **a** indicates the position of **c**, **d**. Coordinates are in UTM zone 46. **e** and **f** are elevation profiles over the WorldView and Copernicus DEMs, with profile locations indicated in **a** and **d**. PlanetScope images courtesy of Planet[73].

Discussion

The evolution of the 2022 detachment of the Bukadaban East glacier underscores that such glacier detachments appear closely linked to surge-like flow behaviour. The event occurred in a region and mountain range with many observed surges. The glacier itself exhibited surge-like behaviour before its detachment, analogous to its western twin, which concluded such activity with a classical albeit sudden and rapid advance. Fine sediments, proposed to have been involved in many other glacier detachments[11] and surges[36], are not visible in any of the satellite images shortly after glacier detachment. We cannot completely exclude that they played an important role in reducing basal glacier friction, but they seem at least not to have been mobilized by the detachment in large amounts.

Instead, the Bukadaban glacier collapse appears to be the result of polythermal glacier conditions (a thermal pattern of zones of temperate (thawed) ice at the melting point and zones of cold ice below the freezing point), where a surge-like mass transfer exerted stress on a resisting glacier terminus that was frozen to its bed to an extent that prevented stress accommodation through terminus advance[37]. The permafrost conditions around the glacier tongue, and a vertical up to 30 m high side wall of the detachment (Fig. 2; “Methods”), support our suggestion that frozen zones have likely existed at the glacier bed, at least at the slow-moving and thin glacier margins. From our data material, though, we cannot definitively conclude the cause of increasing friction loss above the glacier terminus. One scenario of friction-reducing processes is increasing water pressure under and in the glacier (hydraulic mechanism), and associated progressive thaw of the frozen glacier tongue (thermal mechanism). A second scenario is damage accumulation within (frozen) sections of the glacier bed towards failure (mechanical weakening). The exceptional warm summer 2022 in the study area and the likely associated relatively high amounts of glacier melt water produced (Fig. 7) support a hydraulic/thermal process. Indeed, a pond at the western margin of the glacier, and possibly small short-lived blueish features on the glacier surface that could have been supraglacial ponds, very uncertain though, during the weeks and days before failure (Fig. S8), would suggest the existence of water in the glacier even during October 2022, under negative outside temperatures. On the other hand, we could not observe direct signs of large amounts of liquid water on the failure surface after the event, both not on optical nor radar images (Figs. 2 and S12, “Methods”). The light-blue ice observed at the glacier margins during the days before failure (Figs. 5 and S8) could have been (dry) crushed ice or (wet) slush ice, supporting either of the proposed failure mechanisms. The fine ice debris all over the failure surface (Fig. 2) could point to mechanical failure within a (possibly frozen) layer at the glacier bottom, but could also stem from remaining slush ice that quickly froze at outside temperatures of −10 °C and below, or fallout from the detaching mass (Fig. 7).

Whereas mechanisms that involve liquid water are established to explain acceleration and destabilisation of glacier flow, or even

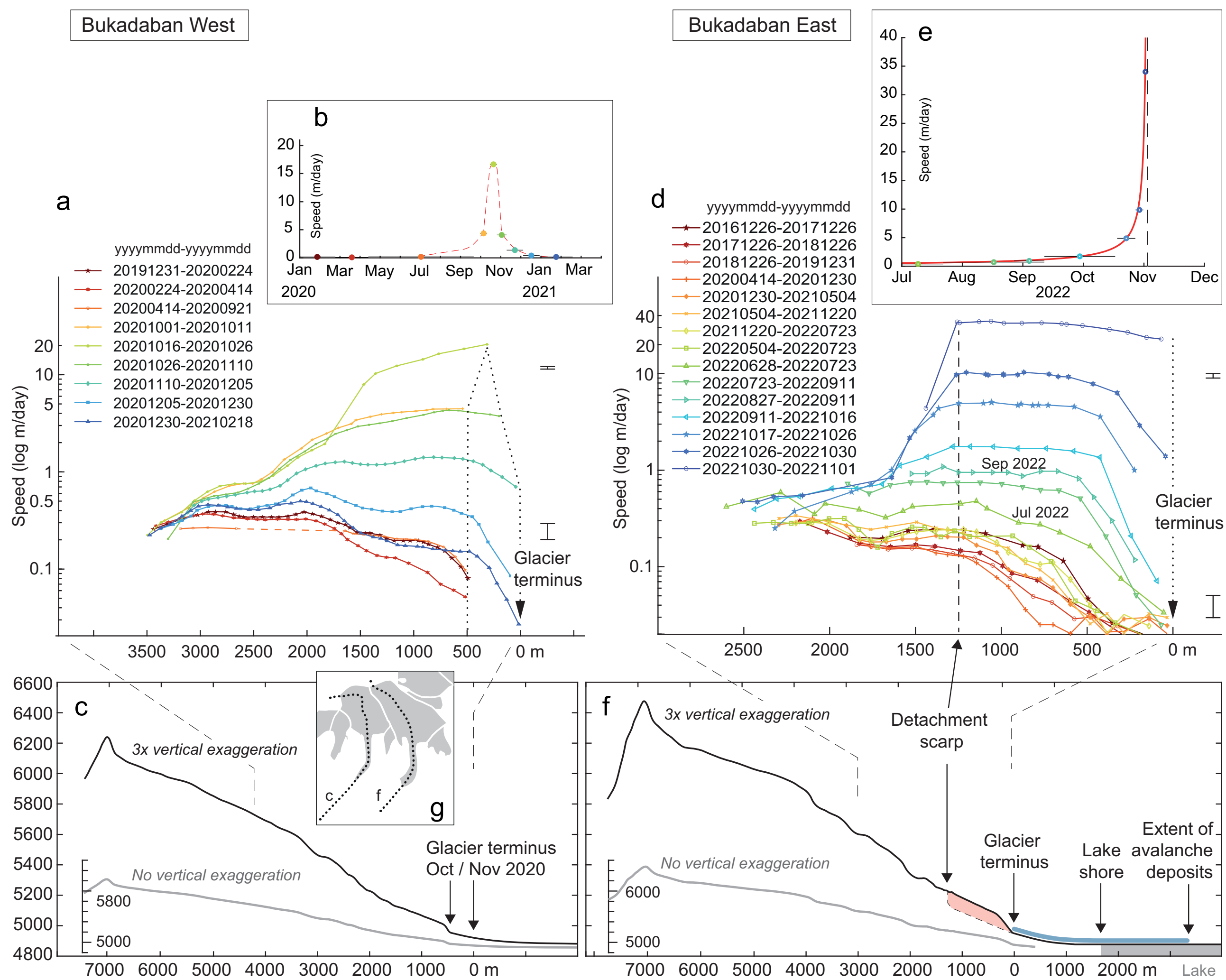

Fig. 6 | Time series of glacier speed along central flowlines and topographic profiles for the surging Bukadaban West glacier (left) and the detached Bukadaban East glacier (right). a Glacier speed time series along a central flowline (location indicated in **g**) of the tongue of Bukadaban West glacier measured by automatically tracking surface features within repeated Sentinel-2 and PlanetScope optical satellite images. Profile section with speed measurements indicated by dashed lines to (**c**). Note, vertical axis in logarithmic scale. Error bars to the right are average stable-ground root-mean-square errors. Size of error bars varies due to logarithmic speed scale and is given for two selected speed levels. **b** Average ice speeds over time for a zone in the middle of the Bukadaban West glacier tongue. *X*-axis: months, *y*-axis speed in m per day. Horizontal bars indicate the measurement periods. **c** Topographic profile of Bukadaban West without and with 3x vertical exaggeration, interpolated from the Copernicus DEM (courtesy EU Copernicus). **d** Glacier speed time series along a central flowline of the tongue of the Bukadaban East glacier using feature tracking in repeat Sentinel-2 and PlanetScope optical satellite images. Profile section with speed measurements indicated by dashed lines to (**f**). Note, the vertical axis in a logarithmic scale. For error bars, see description of (**a**). **e** As **b** but for Bukadaban East, with a power law function fitted[4] (red). The vertical dashed line is the detachment date, 1 Nov 2022. Note, the time scale (x-axis) is different between **b** and **e**. **f** Topographic profile of Bukadaban East without and with 3x vertical exaggeration, interpolated from the Copernicus DEM (pre-detachment). Detachment, avalanche deposits, and lake are only schematically indicated. **g** Indication of central profiles used for **a**, **c**, **d**, **f**.

detachment[12,37], mechanical damage is less. Progressive ice damage has been suggested as mechanism for the failure of steep glaciers[4], and crushed ice is known to develop at the base of ice avalanches and strongly reduce basal friction[16,19]. Both an highly water-saturated ice mass and an extensive layer of crushed ice at the base of the avalanche could, together with the permanently frozen ground, and mostly likely already frozen permafrost active layer, explain the low erosion by the avalanche in the slope between glacier terminus and valley floor, and the fact that large blocks of glacier ice remained intact during their ride (Fig. 2). This limited erosion is remarkable given that a volume of 40×10^6 m^3 rushed over this terrain with speeds of at least tens of km h^{-1}. To our best knowledge, the process of extensive cumulative ice damage has not yet been considered for creating a layer of crushed ice at a low-angle glacier base and remains, therefore, to be investigated in detail. Both highly water-saturated ice or widespread crushed ice at the glacier base could also explain the exceptionally low initial Coulomb friction required to let the avalanche simulation reach the observed avalanche extent. Such a low friction coefficient is typical for hyper-concentrated flows. Strain heating associated with the high ice velocities during the days before failure could also have played a role in destabilisation through reducing ice viscosity and creating liquid water[38,39], in particular if the ice deformation was concentrated in a thin basal layer ("Methods").

Eventually, a spatio-temporally varying combination of hydraulic, thermal and mechanical processes could have led to glacier failure. The exceptionally warm and wet conditions of summer 2022 might well have contributed to or triggered the failure, even if the detachment itself happened in the cold season. Exceptional amounts of water and heat advected with it, at the glacier bed, could have initiated the glacier's acceleration phase to a degree that the termination of water input could not stop the friction loss and acceleration anymore, essentially setting in motion a positive feedback mechanism. While hydraulic/thermal processes might have governed the

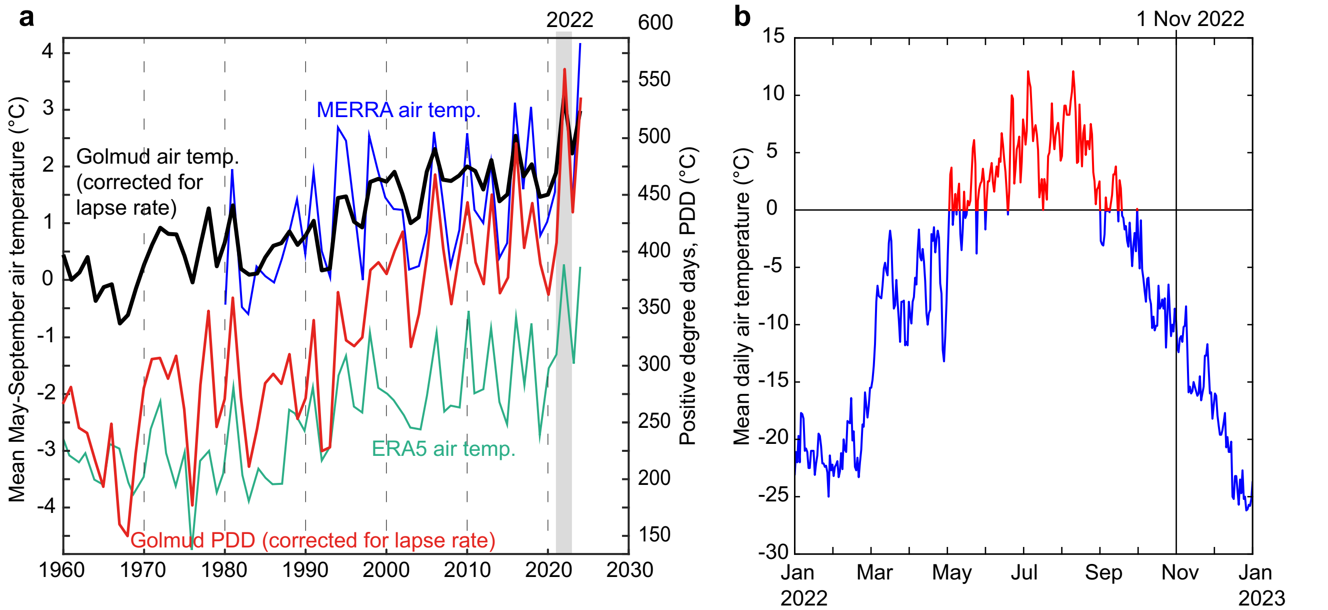

Fig. 7 | Meteorological data and reanalyses, and positive degree days. a ERA5 (turguoise line) and NASA MERRA (blue line) reanalysis average annual May–Sept temperatures interpolated at point 36°N 90.7°E, retrieved from https://climatereanalyzer.org. Average May–September temperatures at Golmud, meteorological station (2809 m asl.), reduced by 14.3 °C temperature lapse to the elevation of the Bukadaban glacier tongues (black line). Temperature data retrieved from https://www.ncei.noaa.gov/cdo-web. Positive degree days (PDD) from temperature data at Golmud reduced by 14.3 °C elevation lapse. The year 2022 had exceptionally high positive degree days (red line). **b** Average daily temperatures at Golmud, reduced by 14.3 °C temperature lapse. The data suggest the detachment site was exposed to consistently negative air temperatures (blue) for about 1.5 months before the event, and around −10 °C for the entire month before the event.

glacier acceleration above the terminus, the failure of the frozen tongue, acting as hydraulic and mechanical dam to the mobilizing mass above, could have been of mechanical nature. It is important to note that the surge-like mass transfer that pushed towards the tongue had already commenced several years earlier.

The role of enhanced geothermal heat flux in the Bukadaban event remains unclear and could have contributed to special basal ice conditions, but the glacier tongue did not show particular signs of instability since the earliest satellite images available from the 1970s. It is unclear to us if the fact that the ice-marginal pond (Fig. S8) remained likely unfrozen even under below-zero air temperatures, is due to geothermal activity, hydrologic connection to a subglacial water reservoir, or a chemical composition that reduced the freezing point.

In sum, the development of the Bukadaban glacier detachment is likely the result of a combination of factors and processes, including long-term surge-like motion, polythermal ice, and exceptional warm and wet summer conditions. While the role of frozen glacier margins, and in particular the terminus, is confirmed by the lack of glacier advance and the remaining vertical side wall of the detachment, the relative importance of either surge-like build-up of ice thickness and associated stresses, and summer weather and associated hydrologic and thermal forcing for the acceleration is difficult to decipher.

The very remote location of the 2022 Bukadaban glacier detachment, with lack of in-situ data available, forced us to exclusively rely on data from satellite remote sensing and seismic stations. Our study demonstrates, however, the wealth of information that can be retrieved by combining a large range of very different types of satellite measurements. Still, the general difficulty of remote sensing to resolve subsurface conditions and processes leaves unavoidably uncertainty in our interpretation of the detachment (see "Methods" for an assessment).

Our study suggests that not only the combination of surge-like flow behaviour with particularly soft sediments, but also with polythermal ice structure can facilitate catastrophic failure of low-angle glaciers. Such polythermal distributions within glacier tongues are widespread in several regions globally and constitute temporally variable conditions as frozen ice sections can decrease but also increase, even amid general atmospheric warming[37,40]. The combination of massively increased ice speeds with a stable terminus position persisted in the case of the Bukadaban East glacier for only a few weeks, but nonetheless indicates potential for a short, possibly crucial early warning window should similar cases develop elsewhere.

Methods

Event timing

Close inspection of available optical and radar satellite data revealed that the ice avalanche had not yet happened on 1 Nov 2022, 4:32 UTC (high-resolution ASTER satellite image) and 4:35 UTC (low-resolution Sentinel-3 image), but was already clearly visible in Suomi/NPP/VIIRS low-resolution imagery of the same day, 1 Nov 2022, 6:54 UTC, and then with high resolution on PlanetScope optical and Sentinel-1 radar data of 2 Nov 2022.

Large glacier avalanches can cause measurable seismic signals[12]. In a first step, we used simple seismic analysis to find the event timing. The only significant seismic event on 1 Nov 2022 between 4:30 and 7:00 UTC registered at the surrounding seismic stations (approximately 910–1700 km away; Supplementary Fig. S2) aligns well with an epicentre at the Xinqingfeng ice cap at roughly 5:55 UTC (13:55 Beijing time) (Fig. 1e), which we assume to be the most likely approximate event time. Our first processing of the seismic signals followed Kääb et al.[12] We downloaded broadband seismic records (BHZ) of the surrounding IRIS stations (https://ds.iris.edu), de-trended them, and applied a 33–100 s band-pass filter. Event timing was then roughly estimated by constructing a trend through the signals sorted by epicentre distance (Fig. 1e). Seismic signals of the closest station (Lhasa) were unfortunately not available from the IRIS catalogue for the time frame of concern.

Seismic signal analysis

In a further step after the above simple screening of seismic signals, we performed a more in-depth seismic analysis. Data used were broadband records from IC, II, IU and KC seismic networks (IRIS Data Management Center's web service, https://service.iris.edu/fdsnws/dataselect/docs/1/builder; Supplementary Fig. S3). Their epicentral distance range is between 912 km and 1912 km. For data pre-processing, we first removed the instrument response, mean and the linear trend. Then, we integrated the seismograms from ground velocity to ground displacement, and rotated the horizontal components to the radial and transverse directions for each station. A fourth-order minimum-phase Butterworth bandpass filter with a period range of 33–100 s was applied. Finally, the horizontal components of the seismic waveforms were removed due to their poor signal-to-noise ratio (SNR).

For signal inversion, we use an inversion algorithm[41] that determines the landquake force time history (LFH) using multiple time-dependent forces. The inversion method is based on the block model approximation (constant block mass during movement). This assumption is appropriate as the spatial scale (~1 km) of the sliding block is very small compared to the wavelength of the seismic waves and the distances (912–1912 km in this study) between source and recording stations. Thus, the signal periods longer than 30 s (wavelength > a few 10 km) are chosen for our seismic waveform inversion. The effect of lateral structural variations and attenuation can also be discounted. Different weightings are assigned in the inversion according to the quality (signal-to-noise ratio, SNR) of the observed waveforms. Here, the SNR value is computed from the ratio between the absolute peak amplitude and the whole-trace average of the absolute amplitude. The synthetics are formed using Green's functions computed with a numerical implementation of the propagator-matrix approach[42] based on the 1-D global velocity and anelastic attenuation model (ak135Q, ref. 43). Due to uncertainty in event occurrence time and the assumed velocity model, the synthetic waveforms do not perfectly align with the observed seismograms. Therefore, the data and synthetic waveforms are cross-correlated, and the observed seismograms are shifted to maximize the cross-correlation coefficients (CC), with an allowance of ±10 s time shifts. The adjustment is done for each component individually. The waveform fitness between synthetic and observed seismograms is defined by both the variance reduction (VR) and CC values.

After waveform inversions, the resulting three-dimensional (3D) LFH that acted on the Earth at the centre of the block mass can be obtained with a fitness value of 0.8518 (Supplementary Fig. S4), a maximum absolute force magnitude of 1.3×10^{10} N, and a source duration of 140 s (Supplementary Fig. S5a). Uncertainties in the resulting LFH mainly rely on the data quality of observed waveforms. Our previous studies[41,44–46] demonstrated that only few waveforms with good signal-to-noise ratio (SNR ≥ 3.0) could achieve a reliable solution (waveform fitness > 0.75). The current event has a much smaller landslide volume than, for instance, the Aru cases and only the vertical component provided seismic data with reasonably good signal-to-noise ratio (SNR). Therefore, the inversion was performed using only the vertical component, which may affect the reliability of the inversion results. Assuming a constant value of collapse mass (m) over time, the three-component acceleration time series can be derived by dividing the force results by m. Furthermore, the velocity and displacement of the sliding mass can be calculated from the once and twice integration of acceleration series, respectively. However, there is a trade-off between m and displacement in the aforementioned calculation. With the assumption of a m value of 0.1×10^{11} kg, the maximum velocity of block mass in southern direction was 11.64 m s^{-1}, and occurred at a time of 17.5 s (Supplementary Fig. S5b). We also estimated the horizontal total block-mass displacement (D_e and D_n) of 348.58 m and 891.10 m (Supplementary Fig. S5c), respectively, and the mass collapse trajectory (Supplementary Fig. S5d). In the first 53 s, the block mass moved downhill in southward direction and was then slightly diverted. After 53 s, the block mass moved in southwestern direction (Supplementary Fig. S5d).

The present seismic analysis was conducted independent of our other observations of the detachment and avalanche. Considering the large distance to the epicentre in relation to its mass and the thus limited data quality, the results based on the seismic signals are consistent with the other observations. Disagreement of maximum speeds and trajectory between the numerical avalanche model and seismic inversion can be due to a number of combined reasons. Due to weak SNR, only the vertical wave components could be used in the inversion, reducing reliability. The inversion assumes one constant mass block and refers to its centroid. This assumption is of limited representativeness for the current ice avalanche that, in reality, spread out considerably and lost mass on its travel. In addition, when reaching the lake, parts of the mass likely floated. We also cannot know whether the avalanching mass behaved as one compact volume, detaching and moving at one point in time (which is the assumption for the seismic inversion), or over a longer time interval. In the latter case, the seismic inversion would pick the largest failure volume, if there is a dominant one, or reconstruct some form of temporal average. Modelled maximum speeds are found only for a small area (see "Methods" section on avalanche modelling). We suggest very low friction for the detachment and little erosion in the frozen slope, reducing force transfer to the ground. Over the lake, force transfer was likely even more reduced. In essence, the seismically reconstructed trajectory (Figs. S5d and S10) refers thus mostly to a hypothetical and synchronous mass centroid of the land-borne part of the avalanche.

Volume estimates

As well-resolved digital elevation models before the Nov 2022 event, we used here in particular the European Union Copernicus DEM (from https://panda.copernicus.eu) with data over the site from 2011 to 2012 TanDEM-X acquisitions and the High-Mountain-Asia (HMA) DEM (from https://nsidc.org/data/hma_dem8m_at) with data over the site from 27 Jan 2015 optical stereo images[47] (Supplementary Table S1). A WorldView satellite high-resolution optical image pair acquired 2 January 2023 covers large parts of the detachment and avalanche area. These images were not acquired in stereo mode, rather in cross-track mode, but the different viewing angles over the limited overlap area still enabled us to generate a stereoscopic photogrammetric DEM of just two months after detachment (Fig. 5), using the standard software PCI Geomatics Catalyst OrthoEngine. Glacier thickness models suggest average depths of the detached glacier part of 100 m (ref. 26) and 52 m (ref. 27).

An ICESat-2 laser altimeter track crosses the lower part of the detached glacier part on 17 Oct 2023 (Supplementary Fig. S6), revealing an average ice thickness of around 68 m along the track (maximum 95 m), relative to the Copernicus DEM, and 54 m relative to the High-Mountain-Asia DEM. We checked and corrected for stable ground vertical offsets between ICESat-2 elevations and the DEMs. Between 2 Nov 2022 (first PlanetScope images) and the 2 Jan 2023 WorldView images, several large glacier ice lamellas have fallen or slided into the detachment area (Figs. 2 and 5). Differencing our WorldView-derived photogrammetric DEM from the Copernicus DEM, and neglecting the latter filled-up areas, gives an average difference of 65 m (median 69 m, maximum 90 m) translating into a failed volume of approximately 38×10^6 m^3. Second, comparing the ICESat-2 elevations over the exposed glacier base (ICESat-2 ATL03, 06 and 08 products)[48–50] to the two ice depth models and calibrating these with a scale factor between the measured and the modelled glacier depths gives for both models a detachment volume of ~$43–44 \times 10^6$ m^3 relative to the Copernicus DEM, and ~$34–35 \times 10^6$ m^3 relative to the HMA DEM.

For a third estimate of involved ice volume, we employ specialised sensor modelling and processing of optical stereo data from the ASTER (Advanced Thermal Emission and Reflection Radiometer) sensor[51], as DEMs from standard processing by NASA are not useful for comparably small glaciers such as Bukadaban East. Comparison of the still rather noisy ASTER-derived DEMs for 25 Jul and 1 Nov (pre-detachment) 2022, 19 Oct 2023 (post-detachment), and the Copernicus and HMA DEMs suggests a detached volume of roughly 30×10^6 m^3 with a stable ground-derived uncertainty of $\pm 5 \times 10^6$ m^3 (1 sigma).

Visual interpretation of satellite imagery over several years before detachment indicates that the tongue bulged up (Fig. 5), and we thus assign most weight to the volume estimates based on the Copernicus DEM. The quite precise WorldView and ICESat-2 elevations, glacier depth modelling, ASTER DEMs, and the estimated pre-detachment bulging in combination suggest thus a detached volume on the order of 40×10^6 m^3 with our approximate uncertainty estimate of $\pm 5 \times 10^6$ m^3. The by far largest component of this rough uncertainty estimate is the geometry of the glacier surface just before failure, i.e. how far it bulged up with respect to the pre-event DEMs. Ice flow started to refill the detached glacier part, and between 18 and 20 Oct 2023 (PlanetScope imagery; Fig. 4), a second volume of roughly 10×10^6 m^3 detached, forming a similar but smaller ice avalanche compared to the 1 Nov 2022 one (see section below on landscape changes). The latter secondary avalanche volume was estimated from the horizontal extent of the detached glacier section and above depth estimates.

Ground thermal regime

Snow-free areas in else snow-covered satellite scenes indicate the existence of enhanced geothermal heat flux at an east-west belt where the slope from the Bukadaban East glacier transitions into the flat valley bottom, most likely warm springs (cf.ref. 25). This finding is confirmed by the fact that the 1 Nov 2022 avalanche deposits melted fast at the location of these springs, and ponds of liquid water could be seen even in winter 2022–2023 satellite scenes of an else deeply frozen landscape, with all other lakes covered by ice and snow cover (Figs. 2 and 4). Ice melt was already visible in a Sentinel-1 radar image 30 h after detachment (Supplementary Fig. S12). A fumarole can be recognized in several high-resolution images within GoogleEarth, and even through ice deposits in the 2 Jan 2023 WorldView scene (for location see Figs. 1 and 3). Further, several ASTER night-time satellite thermal infrared scenes of the area show that the zone of suggested warm springs is typically 10–15 °C warmer than the surrounding surface temperatures, and the warmest site around the entire Xinqingfeng ice cap (Fig. 3). No such geothermal activity or melt features can, though, be seen on the icy failure surface after detachment of the glacier (Fig. 2), nor in an August 2025 night-time thermal infrared under ice-free conditions (Fig. 3b). We assume thus that geothermal activity did not play a major role in the detachment but cannot exclude that it could have transformed and weakened basal ice in the long term, and in general be involved in the surge-type glacier activities of the Xinqingfeng ice cap.

A global permafrost model[24] suggests continuous permafrost with mean annual ground surface temperatures of approximately −3 °C around the glacier tongue, pointing to similar ground thermal conditions as found for the Aru glaciers[24,37] that detached in 2016. Along the Tibet highway, some 200–280 km east of the Xinqingfeng ice cap, mean annual ground temperatures modelled by Obu et al.[24] turn out to be on average roughly 1 °C colder than measured in boreholes[52,53]. This agreement confirms the suggested existence of permafrost around the Bukadaban glacier tongues, perhaps slightly warmer than the modelled −3 °C.

The remaining flanks of the detached glacier section are, in their upper parts, very steep. In particular the orographic left flank at the eastern margin appears to be a ~30 m high, vertical wall of ice, or ice-covered moraine material (shadow along the entire detachment in Fig. 2). This wall is well visible in the high-resolution satellite images 2 months after the detachment, but also even three years later, though likely less high by then (smaller shadow). The vertical side-wall and its persistence hint to frozen material and are thus another indication of frozen glacier margin and polythermal ice.

Avalanche and glacier change observations from satellite data

After melt of the Bukadaban ice avalanche deposits in 2023 and 2024, vegetation patches became visible again at several places on the slope below the glacier tongue in infrared false-colour composites and normalized difference vegetation index data, some similar to the ones visible before the event. Some vegetation patches are visible again immediately after melt of the avalanche ice. In addition, the water channel from the glacier terminus towards the lake and most of the micro-topography appear unaffected by the avalanche (Supplementary Fig. S9). Both observations from satellite images suggest little erosion through the ice avalanche. Among the little changed micro-topographic features are also swarms of ridges/stripes in glacier flow or avalanche direction, respectively. It is unclear to us if these ridges are so-called glacier flutes (elongated debris ridges formed by glacier flow over a soft wet bed) or signs of earlier avalanching, as indeed found for some other glacier detachments[11]. In any case, the features must have formed before the satellite image era, in the present case. In glacier flow direction, the glacier bed after detachment exhibited an average slope of 10°, in places up to 20°. Across flow direction, the transverse slopes reach values of ~90° at the vertical side-wall mentioned in the above section on ground thermal regime.

Bukadaban East glacier (GLIMS ID G090868E35998N_R, RGI ID RGI2000-v7.0-G-13-52167) and West glacier (GLIMS ID G090846E36001N, RGI ID RGI2000-v7.0-G-13-52164) both underwent a similar surge-like mass transfer with an elevation bulge of roughly 20 m or more travelling down the glacier tongues as indicated by elevation differences between the 2011/12 Copernicus DEM and the 2015 HMA DEM, and between 2011/12 and 2018 TanDEM-X DEMs (TanDEM-X DEM change maps[54]; Supplementary Fig. S11). During Oct 2020, Bukadaban West suddenly advanced by 500 m, with the terminus position being stable both before and after this sudden advance. In contrast, the tongue of Bukadaban East showed little length change over the recent 50 years; a slight retreat by about 50 m between the 1970s (Corona spy satellite images) and 2008 (Landsat imagery), followed by an advance of the tongue by some 130 m until around 2011/12, and then again retreat by around 90 m until 2022. In sync with the length changes, the tongue showed clear changes in width; the tongue was up to 800 m wide during advance phases, down to 600 m at the same position during retreat phases. During 2011/12, the tongue seems to have had a similar convex shape than the months before the 2022 detachment, so that we view the 2011/12 Copernicus DEM to be a reasonable volume approximation for the tongue before failure. During the above time periods, the other glacier, Bukadaban West, showed pulsating variations with much larger amplitudes than its eastern neighbour: 1970–1987 roughly 400 m advance; 1987–2008 500 m steady retreat, mostly after 2000; Nov 2007 300 m advance; 2008–2020 200 m steady retreat. This all indicates a much more dynamic length response of the western glacier tongue than that its eastern twin exhibited.

Recent changes of the Xinqingfeng ice cap

Glacier changes of the remote Xinqingfeng ice cap are little investigated. The glaciers showed a slight area shrinkage by about 0.1% yr^{-1} over 1970–2018, and a median mass loss of 0.22 ± 0.17 m w.e. yr^{-1} (w.e., water equivalent) over 1999–2012[28]. Zhang et al.[28] found several glaciers surged or had signs of former surge-like behaviour, including both Bukadaban glaciers of concern here, and suggest that polythermal controls are important for surge initiation and recession of the glaciers of the ice cap. Also, Treichler and Kääb[30] found a slight glacier elevation loss in the region on the order of −0.2 m yr^{-1} over 2003–2008 from ICESat, with both accumulation and ablation areas thinning. The endorheic lakes in the region showed a strong growth since the late 1990s and early 2000s, indicating a main influence of increased precipitation[30]. These results are consistent with results from Zhou et al.[31] for 2000–2015, who found a regional glacier mass loss of 0.16 ± 0.05 m w.e. yr^{-1}, and that the strong increases in lake volumes can only to a small percentage be explained by increased production of glacier melt water, but rather by general wetting in the region. Increase in level and volume of the endorheic lakes in the area continued until 2022 and beyond (Wang et al.[32], and https://hydroweb.next.theia-land.fr). The elevation change patterns on the ice cap from global ASTER DEM time series[29] are dominated by influences from surge-like glacier behaviour such as strong thickening of tongues accompanied by lowering in accumulation areas, or vice-versa[31]. For the glaciers without signs of surge-like elevation change patterns, though, slight elevation decreases of a few dm yr^{-1} dominate over 2000–2019.

Ice velocity time series

In order to investigate the pre-detachment glacier dynamics of Bukadaban East, we measure horizontal surface velocities along a centre-line profile for a large number of time periods using standard cross-correlation[35,55,56] between repeat Sentinel-2 and PlanetScope optical images (Fig. 6d; note the logarithmic scale for the speeds). Until about June 2022, the glacier tongue had background speeds of up to around 0.2 m day^{-1} (70 m year^{-1}) with some minor variations over time. Notably, along the lowermost 500 m of the horizontal profile above glacier terminus, speed was within the measurement accuracy, i.e. below roughly 10 m year^{-1}. From July 2022 on, speed accelerated (Fig. 6), to up to 26 m day^{-1} between 30 and 31 Oct, and 46 m day^{-1} between 31 Oct and 1 Nov (3:42 UTC, about two hours before detachment), and up to 36 m day^{-1} when measured between 30 Oct and 1 Nov 2022. We detect no statistically significant displacement between a PlanetScope image (1 Nov 3:42 UTC, 3 m resolution) and an ASTER image (1 Nov 4:32, 15 m resolution), indicating that speeds in this period must have been below about 100 m day^{-1}.

In distinct contrast to its eastern twin glacier, the advance of the surging Bukadaban West seems to have accommodated the surge-related stresses on the terminus, and speeds quickly fell back to background level.

A power law function, as suggested for ice break-offs from steep glaciers, can be fitted to the speed time series of Bukadaban East glacier (Flotron fit; Fig. 6e)[4]:

$$v = v_o + a \cdot (t_c - t)^m \tag{1}$$

where v is ice speed at time t, v_o the background ice speed, t_c the critical time of failure, and a and m power law parameters. Our speed data are not dense enough over the weeks before failure for solving for all parameters, and we neglect $v_o = 0$ and prescribe $t_c = 2022.83$ (1 Nov 2022). Good fits are then obtained for a of around 0.2 and m around −0.9.

Meteorological conditions

From snow cover and lake/pond ice changes observed in satellite imagery, we conclude that the melt season 2022 started in May (cf. Fig. 7). First thin snow occurred again beginning of October, when also the average daily temperatures suggested by the Golmud station (corrected for elevation lapse) dropped below 0 °C, and soon after lakes started to freeze. According to the ERA5 reanalysis[57], June and July 2022 were rather warm, May and August 2022 were even exceptionally warm (both 3.8 °C above the 1991–2020 normal), and September and October rather normal (May–September 2022 2.1 °C over the 1991–2020 normal of −0.8 °C; 2 m temperature; NASA MERRA[58] gives similar results; https://climatereanalyzer.org). As annual total, 2022 was almost the warmest year for the study region according to ERA5 (1.3 °C above 1991–2020 normal of −10.1 °C; only 2016 and 2021 similarly warm; Fig. 7). MERRA May–September 2022 precipitation was also rather high (40 mm over the 1991–2020 normal of 140 mm; 150 mm over the normal of 400 mm from ERA5). Also at the closest meteorological station, Golmud, 370 km east of Bukadaban Feng, 2022 as a whole and May–September were both exceptionally warm. Mean annual air temperature 1991–2020 at the detachment site is estimated to be around −8 °C (ERA5 − 10.1; MERRA − 6.8; Golmud station, 2809 m asl., corrected for a lapse rate of 0.65 per 100 m: −7.7). Also, positive degree days (cumulative sum of number of days with positive average temperatures times these positive degrees per day) for 2022 were exceptionally high (Fig. 7)[53].

Numerical model of the avalanche

We use the RAMMS::Rock/Ice model[59–61], designed for thermo-mechanical simulations of rock-ice avalanches, to simulate the Bukadaban ice avalanche dynamics and deposits using several scenarios of varying parameters for friction and sediment entrainment, i.e. ground erodibility. We achieve a best fit between simulated and actual avalanche extent when assuming no erodibility in the slope above the lake plane (motivated by a high probability of frozen ground as well as preserved vegetation patches and micro-topography after the avalanche), erodibility on the flat lake shore, where we suggest the existence of warm springs, and high erodibility of the sediments in the lake. The simulated avalanche reaches the observed extent only when a very low initial Coulomb (i.e. dry) friction of 0.05 is used. (Supplementary Table S2). Such a value is in the range of friction parameters typically suggested for hyper-concentrated floods with water contents of >40%[60]. A possible explanation for this low initial friction is thus high water content and pressure within the glacier. Another, less established explanation can be local damage accumulation at the glacier base, transforming the glacier flanks and bed into a layer of crushed ice. The average friction coefficient from avalanche source along the slope until the lake is 0.11, a value very similar to the ones suggested for the Aru collapses[12,23] (between 0.10 and 0.14).

For the avalanche simulation that best fits its real extent, the employed model settings suggest avalanche speeds of up to 200 km h^{-1} (Supplementary Fig. S10). The very high speeds of up to 200 km h^{-1} occur where the terrain is steepest, just below the former glacier tongue (up to 25° terrain slope). The simulated velocity is about 90–150 km h^{-1} where no erosion takes place and where slope angles amount up to ~10°, and 20–40 km h^{-1}, up to 60 km h^{-1}, where erosion occurs, and the slope angles are 4° or less. Modelled maximum ice deposit thicknesses of 15 m (Supplementary Fig. S10) and their distribution fit very well to the elevation differences between the 2 Jan 2023 WorldView and Copernicus DEMs (Fig. 5), and an ICESat-2 elevation profile of 18 Jan 2023 (Supplementary Fig. S6).

Interpretation of failure development

The relatively high temperatures and precipitation amounts (likely mostly rain according to satellite images) for 2022 over the Xinqingfeng ice cap could have contributed to triggering the evolution of the flow instability leading to the Bukadaban East detachment, not least through production of relatively high amounts of melt water (Fig. 7). The high 2022 summer temperatures have on average risen above the freezing point, while the 1991–2020 normal average summer temperatures were below freezing point. At the Nov 2022 detachment surface, though, we find no signs of liquid water, wet surfaces, or unfrozen sediments in the satellite images (Fig. 2; Supplementary Fig. S12). If the glacier bed was thawed and wet directly after failure, it must thus have refrozen within a day or two. Still, friction decreases by high pressure of water trapped behind the frozen terminus, and hydraulic and thermal propagation and enlargement of zones of low friction at the glacier bed[37] towards and into the frozen terminus are clearly a possible—and established—mechanisms involved in the present failure evolution.

Another process that might have contributed to reduction of friction is strain heating. Strain heating (also called deformational heating) and frictional heating are frequently discussed as a potentially important thermal mechanism involved in the surging of polythermal glaciers[62–64]. In the case of Bukadaban East, considerable strain must have acted within the tongue that could have produced enough heat equivalent to the energy needed to melt tens of grams per kg ice or to heat the ice, and thus have reduced ice viscosity considerably[38,39,65,66]. Thereby, a critical assumption is the thickness of the deforming basal layer. Following basic formulas for strain heating of ice[38,39,67] gives for the Bukadaban East glacier tongue and a velocity of 20 m day^{-1} occurring in a basal layer of ice of 5 m, for instance, the production of the amount of energy needed to melt 3–4 kg of ice of 0 °C per m^3 and day, or to heat this m^3 of ice by ~0.5 °C per day (assuming ice just below melting point and perfect immediate diffusion). The latter ice heating has the potential to increase the ice flow rate factor by a factor of ~1.3 per day[67]. For the same setting but a 1 m thick deforming basal layer the energy produced by strain heating could be equivalent to melting 20 kg ice per m^3 or warming ice by ~3 °C per day, with an increase of the ice flow rate factor by 3–4 times. In sum, in particular if a deforming basal ice layer was thin, strain heating can be expected to have played a non-negligible role in the increasing acceleration of ice flow and thus evolution of the detachment.

The facts that the terminus position of Bukadaban East did not change during the build-up of the instability and that the ice speeds on the lowermost horizontal 500 m of the glacier were very low before June 2022, and only slightly increasing during late summer 2022, suggest that the lower part of the glacier was at least in large parts frozen to its bed. This conclusion is supported by the continuous permafrost conditions around the glacier, the low average air temperatures, and thermo-dynamical models of the Aru glaciers (similar elevation as the Bukadaban glaciers) that show frozen glacier termini before their detachments[37]. The 30 m high, persisting vertical sidewall after detachment further suggests the existence of frozen glacier margins.

In contrast to Bukadaban West and its pulse-like advances in 2007 and 2020, the tongue of Bukadaban East seems to have kept up resistance against the increasing stress from the surge-like acceleration from above. The basal ice, at least under the frozen tongue, could have weakened by localized damage accumulation until catastrophic failure. Besides quickly refreezing slush ice, the suggestion of basal ice damage is one possibility to explain the fine-grained ice debris visible on the post-event high-resolution WorldView satellite images from 2 January 2023, in which at least no ice blocks larger than the image resolution of 0.5 m can be observed.

The light-blue ice at the glacier tongue margins, developing before the failure (Fig. 5 and Supplementary Fig. S8), could be dry crushed ice or wet slush ice. Some of the light-blue ice features could also be frozen supraglacial lakes. Certainly open to discussion, we favour, after extensive visual examination of satellite image sequences, the interpretation that most of the light-blue ice features are likely crushed ice, pointing to some ice damage development, at least at the glacier margins. We favour this interpretation over wet slush ice for a number of reasons; the pattern of the blue ice seems rather stable over time, also in steeper glacier parts, not producing flow features. The air temperatures during the occurrence of the light-blue ice must have been around −10 to −20 °C (mean daily temperatures; inferred from the Golmud station; Fig. 7), and surface water bodies were frozen, except the ice-marginal pond (Supplementary Fig. S8). We find similar ice colours in close-by situations of cold-season ice falls and smaller ice avalanches, where the occurrence of slush ice is unlikely.

The lower tongue of Bukadaban East was wider and less laterally constrained than that one of Bukadaban West, thus likely more shallow (as also confirmed by the ice thickness models) and more likely frozen to the bed. The tongue of the Bukadaban East glacier was thinner than the tongues of the Aru glaciers[37], which, together with the very low ice speeds, should have led to larger zones of frozen glacier bed than for the Aru glaciers. Remarkably, no fine sediments were visible on the failure surface and the ice avalanche deposits, in contrast to many of the dozen of other known large-volume detachments of low-angle glaciers[11]. Discovery of Early-Pleistocene fluvial-lacustrine strata ca. 60 km north of the Xinqingfeng ice cap[68], and the variable surface lithology around the ice cap that is visible in short-wave infrared satellite image composites, both suggest that the existence of soft sediments can, though, not be ruled out.

While we find the interpretation of the Bukadaban detachment as caused in parts by cumulative damage of polythermal glacier ice, in particular where frozen to the glacier bed, a possible interpretation that reconciles many of our observations, we cannot exclude other evolutions, as no in-situ observations of the glacier shortly before and after failure are available. The other potential failure development, inspired by the 2016 Aru failures, would involve the accumulation of large amounts of meltwater during a particularly warm summer. Note that we consider the initiation of the accelerated ice velocities by such process well possible in any case. The bluish ice at the glacier margins could have been wet slush ice pressed from the glacier bed to the margins, quickly freezing once reaching the cold outside temperatures. The failure would then have been the detachment of a glacier bed under an enhanced water pressure that exceeded the local stability threshold. Eventually, both hydraulic/thermal and ice damage processes together might be involved in glacier detachments, though with varying relative importance at different times of failure development. In the Bukadaban detachment, zones of frozen glacier bed and basal and marginal ice damage seem to have played an important role in any case.

Despite resembling each other overall, detailed investigations[37] exhibited clear differences in spatio-temporal development of friction, stress and velocity fields between the two Aru glacier collapses in 2016. Neglecting the notable differences between the Aru and the Bukadaban detachments, the Bukadaban event shares larger similarities with the second Aru detachment of 21 Sep 2016 (Aru-2), such as the short-term increases in speed-up (Aru-1 velocities increased over the years) and the stagnant terminus position before detachment (Aru-1 advanced). For the Aru cases, it was modelled that mainly the lateral friction held the ice masses in place prior to detachment[37]. The ice crushing at the margins of the Bukadaban East glacier tongue points also for the present case to high lateral friction, but resistance by the frozen front must have played a crucial role in withstanding the enormous forces associated with a 40×10^6 m^3 ice mass, moving by up to 40 m day^{-1}. This resistance could have contributed to let friction below the glacier even more decrease than in the Aru cases.

Landscape changes

The 2 Jan 2023 WorldView DEM and the ICESat-2 satellite laser altimetry tracks provide the only precise vertical information after detachment (Fig. 5 and Supplementary Figs. S6 and S7). We use ATL03, 06 and 08 products (single photons, land ice heights, land heights, respectively)[48–50]. At the latitude of Tibet, ICESat-2 tracks do not repeat, and only a few ICESat-2 overpasses and their respective beams cover terrain of interest for this study. A track of 17 Oct 2023 over the detached glacier tongue was used for volume estimates (see above). The WorldView vs. Copernicus DEM differences and the ICESat-2 elevation data from 18 Jan 2023 confirm that no noticeable erosion and deposition happened on the slope between the detached glacier tongue and the valley floor, and show ice deposit thicknesses in the valley floor of up to 15 m, at one location even 20 m, well in line with the avalanche simulations (Fig. 5 and Supplementary Figs. S6 and S10). Temporary, two lake parts existed in autumn 2023 (ICESat-2 track 11 Aug 2023), separated by a ring of not yet melted ice avalanche deposits (Fig. 4 and Supplementary Fig. S7). The level of the inner lake basin, newly excavated by the avalanche (see the following paragraph), was by then approximately 1 m higher than the outer lake part that existed already before the detachment, indicating that the ice deposits formed a dam between the two lake parts.

The Nov 2022 ice avalanche led to a massive landscape change in its runout zone through erosion and transfer of sediments (Fig. 4). Roughly 0.5 km^2 of former land area near the lake shore was eroded away and is by August 2025 submerged. From the lack of ICESat-2 lake bottom returns and visual inspection of multispectral satellite images (including PlanetScope coastal blue channels), the water of this newly formed lake area seems deeper than just a few metres. Assuming, for instance an average water depth of 10 m suggests an eroded volume on the order of 5×10^6 m^3. On the opposite side of the lake, roughly 1 km^2 of former lake area had disappeared by October 2024 and was instead filled by fresh sediments. The lake seems to have been less deep at these places before the avalanche event (suggested by optical satellite imagery and ICESat-2). Assuming, very roughly and certainly open to discussion, a former water depth of 1–5 m gives a newly deposited sediment volume of 1–5 10^6 m^3. In addition to the lake changes and the transfer of several million cubic metres of sediments in and at the lake over a distance of on average about 1 km, the river channels have also changed in the avalanche deposition zone, or newly formed, respectively. In total, around 3.5 km^2 land surface was strongly altered by the avalanche impact (Fig. 4).

By Oct 2024, most of the ice deposits by the avalanche have melted away as suggested by optical satellite images (Fig. 4) and confirmed by ICESat-2 elevation profiles (Supplementary Fig. S7). While such a melt rate might seem fast, it is fully expected. The deposits were widely spread and reached only thicknesses of ~15 m. Assuming a degree-day factor of ~10 mm day^{-1} C$^{\circ -1}$ (refs. 12,69) lets us estimate a potential ice melt rate in the order of 5 m year^{-1}. The melt rate might in reality have been even higher as the ice was crushed into large and small pieces (Fig. 2), enlarging its effective surface area, and only a little was covered by protecting lithospheric material. The strongest melt-enhancing factor, however, was certainly the direct contact of large parts of the ice deposits with lake water. Such contact is well known to be able to accelerate ice melt considerably, not least through thermal convection[70,71].

Data availability

Sentinel-1 and Sentinel-2 satellite data are openly available from the European Union Copernicus Browser (https://browser.dataspace.copernicus.eu). PlanetScope satellite data are freely available through academic programs (https://www.planet.com/industries/education-and-research). The Copernicus global DEM is openly available from European Union PANDA. The High Mountain Asia DEM is openly available from NSIDC. Seismic data are openly available from IRIS. ICESat-2 and ASTER data are openly available from NASA Earthdata (https://earthdata.nasa.gov). Landsat and Corona satellite data are openly available from the U.S. Geological Survey (https://earthexplorer.usgs.gov). ERA5 climate data used are openly available from the Copernicus Climate Data Store; we used the University of Maine Climate Reanalyzer for exploration of ERA5 and NASA MERRA data (https://climatereanalyzer.org). Open meteorological data for the Golmud station were downloaded from NOAA. The ice displacements for the glacier velocity timeseries of this study using image matching and the WorldView-derived elevations are available on Zenodo (https://doi.org/10.5281/zenodo.17754687 https://zenodo.org/records/17754687). The raw WorldView and SPOT6 images are commercial and cannot be made available. Academic programmes exist, however, for purchase or free access.

Code availability

The image matching software used (CIAS) is freely available from https://mn.uio.no/icemass. The avalanche simulation software RAMMS is commercial, but academic programs exist (https://ramms.ch).

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Author contributions

A.K. designed the study, processed data, and wrote the text. J.A. simulated the ice avalanche. D.T. processed ICESat data. L.G. processed ASTER DEMs. W.C. performed the seismic analysis. All authors analysed and interpreted data, and wrote and edited text.

Competing interests

The authors declare no competing interests.

Additional information

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